Severity-controllable pathological text-to-speech synthesis for clinical applications

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***Abstract*— The paper presents a new pathological text-to-speech (TTS) synthesis system that has the ability to control speech severity using latent interpolations. Recognising the difficulty of this task, our work uses a data augmentation technique to generate a single-speaker multi-severity training dataset required for training such a model. Furthermore, we show how x-vectors already contain information about the severity and leverage it as a conditioning variable for the synthesis. Finally, we propose modifications to the GradTTS architecture to enhance the duration modelling of pathological speech. We carry out objective and subjective evaluations to demonstrate that the proposed GradTTS system works well, and produces more natural, controllable, and stable pathological speech samples than the baseline TransformerTTS system.**

***Index Terms*— pathological speech, oral cancer speech, clinical text-to-speech**

I. Introduction

SYNTHESISING convincing pathological speech samples is challenging using current speech technology, however, it holds potential for several applications. One key application is the synthesis of voice samples to mentally prepare patients for anticipated changes in their speech, for example, after an oral cancer surgery. While this need can be partially met with natural recordings from previous patients, natural recordings introduce uncontrollable variables such as differences in speaker identity, language, or regional dialect. These inconsistencies can make it difficult for patients to focus on the specific effects of the pathology itself.

A more effective preparatory tool would utilize a single, consistent synthetic voice to demonstrate a disorder across multiple levels of severity. This standardizes the vocal baseline, allowing the patient to understand the progressive impact of the pathology in a clear and controlled manner, without the distraction of a changing speaker identity. The need for better prognostic information is well-documented; 43% of head and neck cancer patients are not satisfied with information about side-effects, including their future voice [1]. Patients also wish to actively participate in treatment decisions [2] using clear, individualized information [3]. While the ultimate ambition is a system that can personalize a patient's own voice, developing a robust, single-speaker model capable of rendering multiple severities is a critical and foundational first step. Therefore, the goal of this work is to develop such a system, providing a significant improvement over the use of inconsistent natural recordings and paving the way for future patient-centric tools.

Pathological speech synthesis and voice conversion have primarily been studied in previous work as intermediate steps for downstream tasks, such as automatic speech recognition (ASR). For instance, systems have been developed to enhance ASR for hard-of-hearing individuals [4] and dysarthric speakers [5]–[8], as well as to improve spoken language understanding for dysarthric speakers [9]. However, relatively few studies directly evaluate the performance of speech synthesis systems [10], [11].

Moreover, existing methods for synthesizing pathological speech are limited in flexibility, which hinders their clinical adoption. For example, voice conversion (VC) techniques typically require voice recordings, whereas a text-only input would be significantly more practical in clinical settings [10]–[12]. While a new text-to-speech (TTS) method has recently been introduced, it lacks the ability to control speech severity—a feature that is crucial for clinical applications [9].

Our work seeks to fill this gap by introducing a novel, severity-controllable pathological TTS approach. The aim is to build a TTS system that can synthesise pathological speech from a single speaker with any desired severity, based on text and a control variable.

The primary challenge is that it is impossible to collect the appropriate training data for this task, as one would require speech from the same speaker with several different severities. It is also not straightforward how the control of the severity could be achieved. One approach would be using the severity of the speech as a conditioner, however, that requires labelling by speech language pathologists (SLPs).

To address these challenges, we develop a pathological TTS system using the following novel techniques[1]:

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[1] Audio Demos: `bit.ly/path-tts-demo`

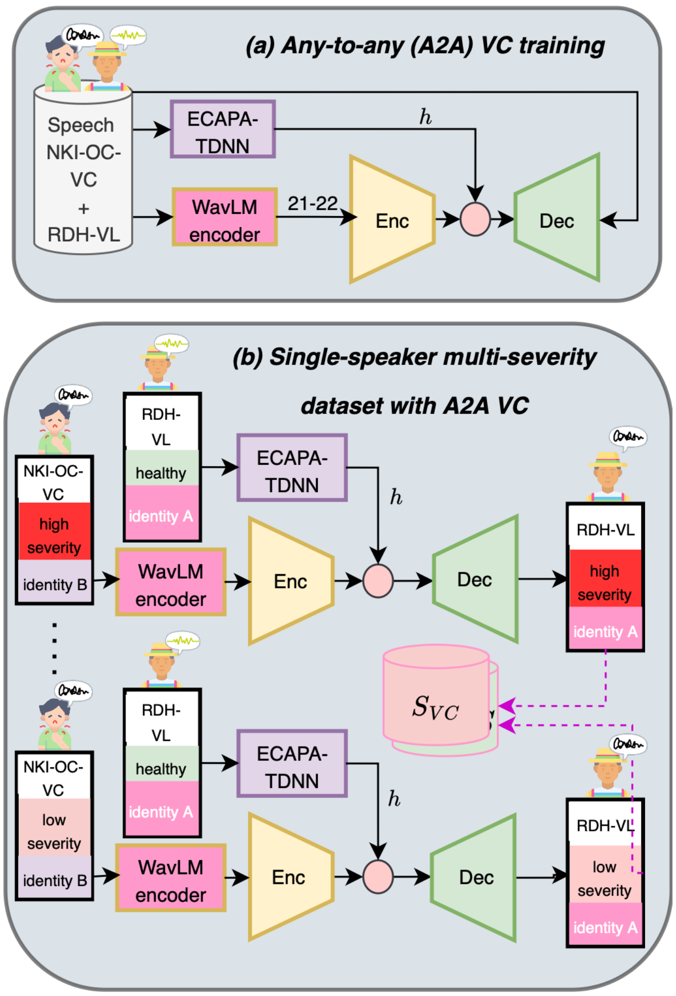

Fig. 1. Outline of the proposed text-to-speech approach. The upper part (a) shows the generation of the VC samples, while the lower part (b) shows how two example (high and low severity) utterances are generated in the single-speaker multi-severity dataset. NKI-OC-VC represents a speaker from the oral cancer dataset, RDH-VL represents the Dutch single-speaker, ECAPA-TDNN stands for the speaker verification model extracting the x-vector, and WavLM21-22 stands for the extracted representation for the conversion.

Data augmentation with multi-severity speech dataset: Building on prior work [11], we introduce a VC-based data augmentation technique to create a single-speaker, multi-severity speech dataset from as little as 2h of pathological speech from different speakers.

Severity control through x-vector conditioning: We use x-vectors to control the severity of the speech [13] which is shown to contain information about the severity, therefore no explicit severity labels are needed for training the system.

Improved duration modelling using GradTTS: By successfully integrating x-vectors into a non-autoregressive (non-AR) GradTTS architecture [14], we enhance its ability to model the duration of speech, which alleviates word-skipping issues of autoregressive (AR) architectures, and creates pathological speech with a more realistic tempo.

We also address the following research questions to validate our proposed approach:

RQ1 Does GradTTS improve stability over TransformerTTS and produce samples with durations akin to those of pathological speakers of similar severity? (**Stability**)

RQ2 Does GradTTS offer more linear control over severity (as quantified by (a) duration and (b) subjective ratings) than TransformerTTS? (**Severity control**)

RQ3 Does GradTTS improve the naturalness of synthesised pathological speech over TransformerTTS? (**Naturalness**)

RQ4 Does GradTTS synthesise samples that are more similar in intelligibility to the pathological speech compared to TransformerTTS? (**Intelligibility**)

Although speaker identity is not typically evaluated for TTS models, our architecture's strong resemblance to multi-speaker systems makes it a crucial aspect to investigate. We therefore pose the following question:

RQ5 Is the developed system indeed a single-speaker multi-severity TTS system or a multi-speaker system? (**Speaker identity**)

II. Severity as a linear construct

One of the underlying assumptions of our work is that severity is a single, simple measure. However, it is a complex concept with multiple dimensions. A clinician's rating of severity is an overall judgment based on many factors, including how clear the speech is (intelligibility), the speaker's talking speed, and the amount of effort a listener needs to understand them [15]. A complete model would ideally account for all these different aspects of speech.

Despite the clinical complexity of severity, its use as a single, simplified measure is commonplace when developing speech technology. For tasks such as automatic severity classification or building adaptive recognition and synthesis systems, severity is often distilled into discrete categories (e.g., mild, moderate, severe) or mapped to a single continuous variable, such as a word intelligibility score [16], [17]. This simplification is often necessary, as it is difficult to create multi-target evaluations.

Our work uses this idea as a starting point. We adopt a simplified approach that represents severity as a single, continuous value. This is a practical choice for two main reasons. First, it allows us to build a system with a straightforward, intuitive control that a user can adjust. Second, the detailed data needed to model each aspect of severity separately is often not available for training such systems. Our method, therefore, combines the multiple dimensions of pathological speech into one variable for the purpose of control.

Therefore, this paper does not claim to capture the full clinical picture of severity. Instead, our goal is to test if this simplified, one-dimensional control can produce useful and predictable changes in synthesised speech. Our research questions (particularly RQ2) are designed to check how well our system's control aligns with what people hear as a logical progression in severity. We believe this is a vital first step, and we consider building more advanced models that can separate and individually control factors like speech clarity and speed to be an important direction for future work.

III. Datasets

NKI-OC-VC: We used the NKI-OC-VC dataset [11], which consists of Dutch pathological speech recordings from 16 oral cancer speakers (10 male) and 5 healthy controls (2 male) who had undergone a composite resection (COMANDO; Combined Mandibulectomy and Neck Dissection Operation) surgery or comparable treatment for mostly advanced tongue tumours. The surgery is an extensive surgery including the mandible, floor of the mouth, and possibly glossectomy. Consequently, the speech impairment in this cohort is predominantly characterized by articulatory (filter) deficits due to restricted tongue mobility and altered oral cavity resonance, rather than laryngeal (source) deficits. The data set includes recordings at three stages: before surgery, immediately, and 6 months after surgery, totalling approximately 2.5h of speech. Recordings were conducted during speech therapy sessions, where subjects read from the standard Dutch text *Jorinde en Joringel* [18]. Due to the variability in patient ability, session lengths varied, with an average of 5 mins per session and 92 sentences per stage per speaker. The speech used was sampled at 16 kHz with 16-bit depth. We divided the 92 sentences from each session into sets for training, development, and test sets (78/7/7 utterances). The dataset includes speech severity labels provided by five speech-language pathologists (SLPs) using a five-point Likert scale with 5 meaning healthy, and 1 meaning severe. The mean severity rating of the raters is used. The raters expertise varied between less than 1 years up to 16 years of experience since graduation.

The interrater correlation between the severity scores was excellent, so the scores are more than reliable for further analysis ((*Intraclass correlation coefficient (ICC) 2,k*)=0.97).

RDH-VL: We also used the Dutch single-speaker RDH-VL subset of the Mozilla Common Voice for our experiments [19][2]. It contains 12.6h of recordings, and $15k$ utterances which we have partitioned into training, development and test sets ($14.5k/250/250$ utterances). The dataset is 22050 Hz and 16-bit which we downsampled to 16 kHz for all of our experiments.

COPAS: COPAS contains 16 kHz audio recordings from 319 Dutch speakers with and without speech disorders [20]. As the NKI-OC-VC cannot be shared publicly, and thus cannot be used in crowdsourcing evaluation studies, we use a small number of utterances from this dataset for our subjective experiments, see Section V-D for more details.

IV. Proposed approach

Our proposed approach to solve the severity controllable TTS task consists of four steps: (1) training an any-to-any severity-preserving voice conversion model, (2) creating a multi-severity single-speaker dataset via inference, (3) training a single-speaker TTS model conditioned on x-vectors, and (4) using interpolation to provide severity control.

A. Training an any-to-any (A2A) severity-preserving voice conversion model

Our approach begins by training an any-to-any (A2A) severity-preserving voice conversion (VC) model, following the methodology outlined in [11] and illustrated in Figure 1 (a). We employ the Taco2-AR architecture implemented in S3PRL-VC[3] [21]. Unlike the phonetic posteriorgram features used in [11], we utilize layers 21-22 of WavLM features [22], [23], as they offer comparable speech quality while eliminating the need for a separate automatic speech recognition model.

To enhance data robustness, we apply a noise augmentation strategy during training, mixing samples from the WHAM! dataset [24] at a fixed 20 dB signal-to-noise ratio. Speaker information is incorporated using the ECAPA-TDNN x-vector model [25], [26], which remains fixed during training. The VC model is trained on the NKI-OC-VC dataset (2h) and the RDH-VL training set (12.3h), yielding a total of approximately 15h of training data (without the augmentations).

B. Creating a multi-severity single-speaker dataset

Once trained, the VC model is used to generate a multi-severity single-speaker dataset, as shown in Figure 1 (b). Specifically, we extract the x-vector centroid $h_{speaker}$ of the RDH-VL speaker and use it to perform voice conversion on all recordings in NKI-OC-VC and RDH-VL. The goal is to preserve the severity variations while preserving the speaker identity. In the upper part of Figure 1 (b), an example is shown with a high severity utterance; and in the lower part an example is shown with a lower-severity utterance. The conversion is repeated for all speakers until obtain the multi-severity single-speaker dataset. We obtain a 15h dataset with a single-speaker but varying severity levels. It is important that during the inference the noise data augmentation is not used. The resulting speech dataset and its transcription is denoted S_{VC} and T_{VC} respectively.

C. Training a single-speaker TTS model conditioned on the x-vector

Using the generated multi-severity dataset, we train a single-speaker text-to-speech (TTS) model conditioned on the x-vector. The training and inference processes are identical for both architectures, as illustrated in Figure 2. It is an encoder-decoder setup where a bottleneck representation $z = f(Y, h)$ is learned, where $Y = \{y_1, y_2, ..., y_t\}$ corresponds to the text, and h is the x-vector representation. After that, h is used again at the decoder level alongside the bottleneck representation as $g(z, h)$ to obtain $X = \{x_1, x_2, ..., x_t\}$, which corresponds to the Mel-spectrogram frames. The repeated concatenation has the role of injecting the severity information into the time-variant and the time-ivariant levels of the speech signal.

The following two architectures only differ in the details of the encoder-decoder mechanisms, and are compared which are compared side-to-side in Figure 3 and described below.

TransformerTTS: As a baseline, we use the default x-vector-conditioned TransformerTTS [27] ESPNet recipe [28].

[2]https://github.com/r-dh/dutch-vl-tts

[3]https://github.com/unilight/s3prl-vc

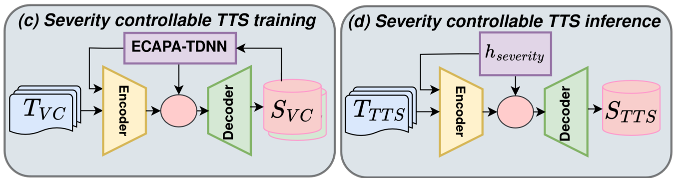

Fig. 2. General outline of the training (c) and inference (d) with the severity controllable TTS models. T_{VC} stands for the text used for voice conversion, S_{VC} stands for the speech used for voice conversion, T_{TTS} stands for text for text-tos-speech, S_{TTS} stands for speech for text-to-speech.

The model effectively tries to learn the autoregressive (AR) probability distribution of

$$P(X|Y) = \prod_{t=1}^{T} P(x_t|x_{<t}, Y, h). \tag{1}$$

The AR model generates each frame based on prior outputs, which can be limiting for pathological speech due to its highly variable phoneme durations. Because the actual sounds often differ significantly from the intended sounds in pathological speech, our preliminary experiments indicated that graphemes demonstrated greater stability than phonemes.

An additional challenge of AR models with using pathological speech is the stop token prediction. For example, stuttering exhibits an arbitrary number of repetitions in sounds, which can result in poor stop token prediction, leading to stuttering-like behaviour in cases where it is not expected at all.

GradTTS: Our proposed model is based on GradTTS [14], a non-AR approach that predicts phoneme durations using monotonic alignment search. We follow the parameters of this implementation[4] with the difference that the x-vector representation h is concatenated before the encoder. This allows the duration predictor to learn an explicit relationship between the severity and the phoneme durations, which is usually an increased phoneme duration. This makes it more robust to the duration variability in pathological speech.

The x-vector h is also incorporated in the decoder as,

$$X_0 = s_\theta(X_T, \mu, t, h), \tag{2}$$

where X_t is the stochastic diffusion process at time t, which is successively transformed by the decoder network s_θ, θ is the parameter set to optimise, and μ are the frame-wise features output by the aligner.

Vocoder: For both models, HifiGAN [29] is used as the neural vocoder to transform the generated mel spectrograms into audio waveforms, trained on the RDH-VL dataset, with implementations sourced from [5]. All models were trained using an NVIDIA GeForce RTX 2080 TI GPU.

D. Interpolating the x-vector for severity control

To enable severity-controllable synthesis, we use x-vectors as a conditioning variable. X-vector is used for the task of automatic speaker verification. However, the feature has been

[4]https://github.com/huawei-noah/Speech-Backbones/Grad-TTS
[5]https://github.com/kan-bayashi/ParallelWaveGAN

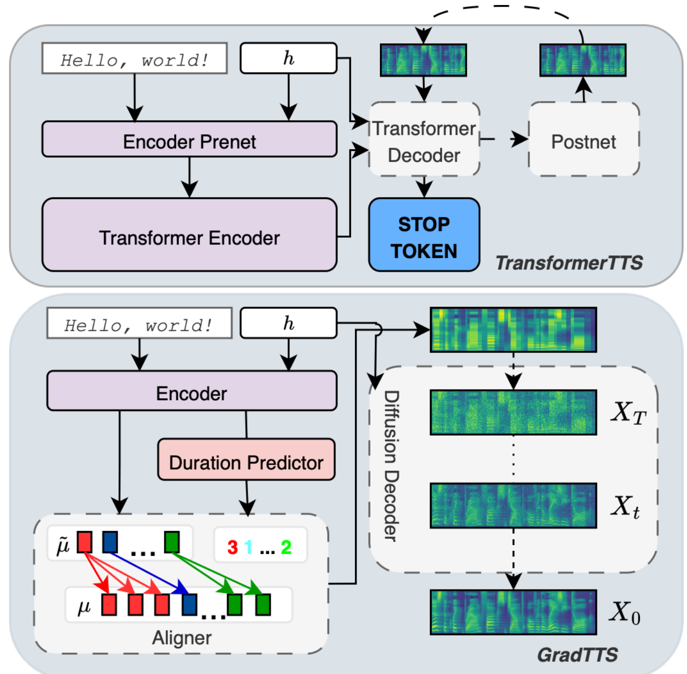

Fig. 3. Comparison of TransformerTTS (up) and GradTTS architectures (bottom). GradTTS incorporates an explicit duration predictor, making it more robust to the timing variability in pathological speech. The notation h is the x-vector representation during training, and the interpolated $h_{severity}$ during infernce.

found useful in several applications including speech severity evaluation [17], detection of Parkinson's speech [30], pathological speech detection [31], [32], speech intelligibility evaluation of patients with total and partial laryngectomy [33], speech severity evaluation of oral cancer speech [34]. It has also been shown that x-vector is sensitive to pathological speech characteristics, and in a different way than humans would be when comparing similarity of two speakers [35]. It is likely when distinguishing different speakers, vocal qualities like whisperedness, and nasality are learned, which the x-vector is thought to be sensitive to [36].

Severity control at inference is achieved via linear interpolation between the x-vectors of healthy and pathological speech:

$$h_{severity} = h_{healthy} \cdot x_{sev} + h_{pathological} \cdot (1 - x_{sev}), \tag{3}$$

where $x_{sev} \in [0, 1]$ is the severity interpolation variable, $h_{healthy} \in \mathbb{R}^{192}$ is the centroid x-vector of the RDH-VL speaker, and $h_{pathological} \in \mathbb{R}^{192}$ is the centroid x-vector of the most severely impaired speakers.

The resulting vector is used for inference at both the encoder and decoder level as $X = g(f(Y, h_{severity}), h_{severity})$.

Unlike traditional supervised approaches, our method requires only extreme severity labels, making it an almost-unsupervised approach.

E. Empirical justification for x-vector severity control

To empirically validate the use of x-vectors for severity control, we conducted a principal components analysis (PCA) on the x-vectors extracted from the NKI-OC-VC dataset. We

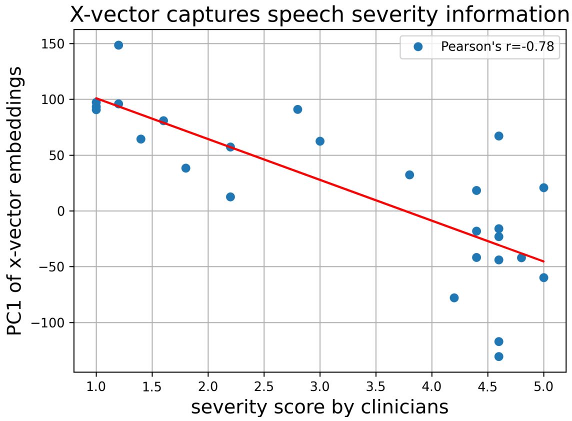

Fig. 4. Scatter plot showing the correlation between the first principal component of x-vectors on the dataset and severity labels.

then computed the correlation between the first principal component (PC1) and the severity labels assigned to the dataset. As shown in Figure 4, we observed a strong correlation ($r = 0.78$) between PC1 and the severity labels, indicating that x-vectors inherently encode severity-related speech characteristics. This high correlation suggests that interpolating between extreme severity x-vectors provides a meaningful and continuous way to control speech severity in the synthesized speech.

V. Evaluation protocols

A. RQ1 and RQ2(a): stability and control

To assess the duration stability (RQ1) of our system, i.e., that the synthesised speech has more realistic durations, we can compare the durations of the synthesised samples with the durations of the real pathological samples. Specifically, we compare the mean durations of real pathological samples with synthesised samples. We synthesise the test set samples of the NKI-OC-VC dataset using both systems, setting the interpolation variable x_{sev} from values of the set $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ to systematically vary the severity, and thus the duration. Then, we compare the linear fit of the mean durations from synthesised samples and the mean durations from NKI-OC-VC the test set using the root mean squared error (RMSE) measure. A lower RMSE means a closer correspondence to the real durations.

To assess the control of the severity of our proposed system using objective measures, we can exploit the phenomenon that pathological speech, in general, exhibits a reduction in speech rate, leading to an increase in duration as the speech becomes more severe [37]. This has been shown many times for many speech pathologies. In the case of laryngeal cancer, a strong correlation ($r = 0.60, p < 0.01$) was reported between intelligibility and speech rate [38]. In the case of a large ($n = 136$) head and neck cancer dataset, also strong correlations were observed between speech rate and intelligibility ($r = 0.68$), voice quality ($r = 0.69$), and articulatory precision ($r = 0.63$) [39]. An exception is dysarthria where both an extremely fast and an extremely slow speech rate is used as a diagnostic criterion [40].

Quantitatively (RQ2(a)), we compare and calculate the correlation between the interpolation variable x_{sev} and the mean

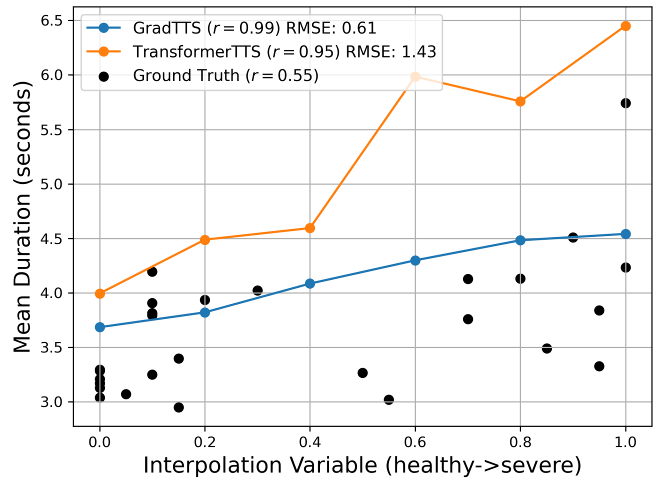

Fig. 5. Mean duration values of the synthetic (GradTTS, TransformerTTS) samples and ground truth samples. r indicates the correlation between the interpolation variable and the mean duration.

durations, a higher correlation meaning a more linear control. As the ground truth samples do not have a straightforward correspondence to the interpolation variable, we assume that the interpolation variables linearly varies between 1 to 5, and we map to the interpolation variable correspondingly. For example, 5 (healthy) corresponds to an interpolation variable of 0, while 1 (severe) corresponds to 1.

B. RQ2(b) and RQ3: severity and naturalness

To evaluate the naturalness of the synthesised samples (RQ3), we carried out subjective listener tests. Naturalness evaluation of pathological speech synthesis is challenging because listeners are known to differentiate poorly between the naturalness and the severity of the speech [10], [11], [41]. To address this, we decided to modify previously used evaluation protocols. First, we combined the naturalness and severity tests into a single questionnaire. This change aims to encourage listeners to give different answers to the questions, enabling us to assess both properties of the developed systems.

Second, we modified the questions. For the severity, we asked the question following [34]. This method has been proven to produce severity scores that are similar to expert listeners but using non-expert listeners. For naturalness, we modify the instruction of [42], emphasizing that skipping/repetition mistakes as a point that should be rated negatively. *A natural utterance is 5, by which we mean that the utterance sounds as if it were spoken by a real person. An unnatural utterance is 1, by which we mean that (1) the utterance sounds as it was spoken by a robot (2) the utterance contains skipped or repeated words and sounds. (3) The utterance contains inappropriate pausing.*"

Following [10], [11], we used a 5-point scale with 0.5 increments to avoid the flooring effect. The stimuli for the experiment consisted of speech samples from $x_{sev} \in \{0, 0.25, 0.5, 0.75, 1\}$ for both systems using 5 sentences from the RDH-VL test set. The 5 sentences from the ground truth (Healthy Ref) and 2 sentences from the glossectomy speaker

TABLE I

RESULTS OF THE SUBJECTIVE SEVERITY AND NATURALNESS EXPERIMENTS BY THE LISTENERS. x_{sev} STANDS FOR THE INTERPOLATION VARIABLE. **BOLDFACE** SHOWS IMPROVED NATURALNESS. NOTE THAT WE EXPLICITLY REFRAIN FROM BOLDFACING SEVERITY AS A HIGHER/LOWER SEVERITY DOES NOT MEAN A BETTER SYSTEM.

	Naturalness					**Severity**				
x_{sev}	1 (Severe)	0.75	0.5	0.25	0 (Healthy)	1 (Severe)	0.75	0.5	0.25	0 (Healthy)
TransformerTTS	1.40±.11	1.52±.12	2.20±.14	2.34±.16	2.83±.17	1.22±.06	1.49±.09	2.36±.14	2.49±.15	3.10±.17
GradTTS	**1.86±.14**	**2.09±.15**	**2.43±.18**	**2.84±.19**	**3.39±.17**	1.49±.09	1.90±.13	2.55±.18	3.39±.17	3.91±.15
Healthy Ref	—	—	—	—	4.53±.12	—	—	—	—	4.71±.10
Pathological Ref	—	—	2.79±.36	—	—	—	—	2.25±.31	—	—

(G1) of COPAS [20] (Pathological Ref) were also included, a total of 57 sentences. Note that ground truth does not exist for some of the samples. For the experiment, we recruited 30 native Dutch listeners (22 male) who do not have hearing difficulties through the Prolific platform. The age range of the listeners was between 19 and 68 years, with the average being 35.1 years.

Severity scores with better correlation to the interpolation variables are considered to indicate better control (RQ2(b)). The system with a higher naturalness score is considered more natural (RQ3).

C. RQ4: intelligibility

To evaluate the intelligibility of the synthesised samples, we evaluate the phoneme error rate for each speaker. We will compare the intelligibility of the synthesised samples to the two extremities using the samples from the best and worst speakers.

To obtain prediction for the phonemes in the utterances, we used a Dutch phoneme recogniser[6] pre-trained on the Dutch Common Voice dataset [19]. All of our utterances had word-level transcriptions, which we converted to phoneme-level transcriptions using the Dutch `espeak` frontend of *phonemizer* [43].

D. RQ5: speaker similarity

To check whether the system is indeed synthesising the voice identity of a single speaker with different severity instead of multiple speakers, we conducted a subjective speaker similarity experiment. The aim of this experiment is not to show superior performance of the GradTTS over the TransformerTTS system but rather to refute the claim that the systems are multi-speaker TTS systems rather than single-speaker multi-severity TTS systems. The basis of this claim could be due to the fact that the x-vector is known as a technique for speaker control [44], and we also use it in our data augmentation to control speaker identity. However, an increasing body of research exploits x-vector representation for speech disorder analysis [17], [30], [31], [33], so in a single-speaker training setting, the x-vector can be used to capture severity.

A straightforward evaluation would be comparison of the synthesised samples to the healthy target speakers, and showing that a high similarity (over 50%) is achieved. However, we expect that the similarity will decrease due to increasing severity (see [10]), therefore we also included a comparison to healthy reference and a comparison to pathological reference. The expectation is that with increasing severity, there will be an increased similarity to the pathological reference, and with decreasing severity, there will be an increased similarity to the healthy reference. The experiment conducted in general is very similar to a voice conversion similarity experiment, however, instead of source samples, we have two (healthy and pathological) references.

The experiment details are as follows. Listeners were asked to rate the speakers on a 4-point scale, responses being: Same, Same (Not Sure), Different (Not Sure), Different, following [42]. We modified the last sentence of the VCC2020 [42] to reflect pathological speech conditions as follows "*Please do not consider the pathology, articulation issues or pronunciation that the speaker might or might not have.*". Despite this instruction, we expect that the listeners will consider pathology as described above.

The stimuli consisted of speech synthesised for 5 different severity levels ($x_{sev} \in \{0, 0.25, 0.5, 0.75, 1\}$ (5 severity levels)) for both (2) systems using 5 sentences from the RDH-VL test set, which were compared to the 5 RDH-VL references, which we call (healthy) synthesis target in the experiment. An additional 2 sentences with the identifier S1 and S2 from a healthy (N57) and a pathological (glossectomy) speaker (G1) from the COPAS dataset [20] were used. Therefore, there are a total of 7 reference sentences compared, across the 2 TTS systems with 5. This leads to a ($(5{\cdot}2+1){\cdot}7 = 77$) comparisons, with the 1 being the reference. Finally, we included a total of 18 comparison for same speaker and different speakers, which we have randomly chosen. Adding this, leads to 95 comparisons in total. For the experiment, we recruited 30 native Dutch listeners (16 male).

To further assess whether x-vectors capture severity variations in the developed TTS system, we conducted an additional visualization experiment comparing the similarity of x-vectors using the entire NKI-OC-VC dataset. To ensure that differences in linguistic content do not influence the x-vector comparisons, we performed only same-utterance comparisons across different speakers. Specifically, for Healthy (control) speaker comparisons, each Healthy (control) speaker's x-vector from the NKI-OC-VC was compared to that of a different NKI-OC-VC Healthy (control) speaker of the same gender. For Pathological speaker comparisons, each Healthy (control) speaker's x-vector was compared to that of a different but same-gender pathological speaker. For the TTS-generated speech, we compared x-vectors to those of all male Healthy (control) speakers, as the VC voice is male. We repeated the TTS comparison with three different severity levels ($x_{sev} = (0, 0.5, 1)$).

If x-vectors are sensitive to severity, we expect a structured similarity pattern, where with the severity control the TTS x-vector distribution gradually transitions from the healthy reference distribution to the pathological reference distribution. Conversely, if x-vectors do not move at all with the severity, it follows that the speaker effects dominate the severity effects in the system overall.

VI. Results and discussion

A. RQ1 and RQ2(a): stability and control

Figure 5 shows that the RMSE of the GradTTS is lower, meaning that the synthesised samples have more realistic durations than the samples by the TransformerTTS (RQ1). Also, the GradTTS system attains a higher correlation with the value of the interpolation variable than the TransformerTTS system, indicating a more linear relationship between the interpolation variables and the mean duration values, therefore a better control (RQ2(a)). It is interesting that the ground truth samples achieve a lower correlation with the interpolation variables. This is expected, as the ground truth comprises multiple speakers, introducing variability in speaking styles and duration patterns. However, this does not undermine the experiment, as the evaluation focuses on the controllability of synthesized systems under consistent single-speaker conditions. The ground truth serves as a benchmark for real-world variability but does not directly affect the comparison of the systems' performance. The large difference between the values of the GradTTS and the TransformerTTS system is due to the poor alignment learning of the TransformerTTS. While it is known that the GradTTS leads to more robust alignments, this robustness difference becomes essential with pathological TTS. This is well illustrated by the duration values in Figure 5. At the lower (healthy) interpolation values, the mean duration of the synthesised speech by the two systems differs by less than a second, while at the higher (severe) interpolation values the difference becomes multiple seconds. We conclude that the GradTTS system is more stable and exhibits a more linear control over the severity than the TransformerTTS.

B. RQ2(b) and RQ3: severity and naturalness

Before interpreting the results, it cannot be emphasised enough that rating the naturalness of pathological samples is difficult. In our subjective experiments, the ratings of naturalness and severity on the same samples exhibited a relatively high ($r = 0.70$) correlation. However, this correlation is lower than the value calculated between the mean naturalness of speakers according to non-expert listeners and severity according to expert listeners ($r = 0.88$) in the work of [11]. Because the main differences between these experiments are the usage of expert listeners for the severity, and the modified rating scheme, we think that the latter is responsible for this reduction in correlation.

TABLE II
Comparison of phoneme error rate (PER%) of the two systems. **Boldface** indicates lower PER.

System/x_{sev}	1 (Severe)	0.75	0.5	0.25	0 (Healthy)
TransformerTTS	102.0	100.3	84.8	80.8	75.9
GradTTS	**89.2**	**86.5**	**78.6**	**65.0**	**63.5**
Reference	100.3				56.3

Table I shows the results of the naturalness experiment (RQ3). In general, the naturalness values reported for the GradTTS system consistently achieve higher scores. The perceived naturalness decreased with increasing severity, which is due to the difficulty of differentiation of these phenomena. GradTTS system are always higher. For RQ2(b), we find that the correlation between subjective severity scores and the interpolation variable is higher for the GradTTS system ($r = 0.9942$) than for the TransformerTTS system ($r = 0.9796$). Along with the evidence presented in RQ2(a), we conclude that the GradTTS system has superior control.

C. RQ4: intelligibility

To address our fourth research question (RQ4) on intelligibility, we conducted an objective evaluation using Phoneme Error Rate (PER). The results in Table II compare our two systems, TransformerTTS and GradTTS, against human speech references.

The results indicate that GradTTS yields consistently lower PER (better intelligibility) compared to TransformerTTS across all severity levels. We attribute this to the superior generative quality and naturalness of the diffusion-based architecture. Additionally, PER increases as x_{sev} increases for both systems, confirming that both models correctly reflect the decrease in intelligibility as pathological characteristics are intensified.

With regards to the similarity of the references, only the references at the two extremities are available for comparison. On the one hand, in the case of the most severe sample, the TransformerTTS (102.0%) is closer than the GradTTS (89.2%) to the reference (100.3%). On the other hand, in case of the most healthy sample, the GradTTS (63.5%) is closer than the TransformerTTS (75.9%) to the reference (63.5%). However, we argue that the apparent alignment of TransformerTTS with the severe reference is misleading. This result contradicts our perceptual severity findings and is likely due to the imprecision of PER when applied to highly unintelligible speech.

In conclusion, we find that GradTTS synthesises samples that are more similar in intelligibility to the target speakers, particularly in the healthy condition where the ground truth is reliable. While TransformerTTS approximates the PER of severe speech more closely, this is likely a conflation of pathological intelligibility loss and synthesis artifacts.

D. RQ5: speaker similarity

The results in Table III confirm our expectations, even in the most severe conditions achieve at least 67% similarity, validating that the architectures are single-speaker TTS. As

TABLE III
RESULTS OF THE SUBJECTIVE SPEAKER SIMILARITY EXPERIMENT.

Comparison to (healthy) synthesis target						
x_{sev}	1 (Severe)	0.75	0.5	0.25	0 (Healthy)	Mean
TransformerTTS	81±4%	83±5%	86±4%	90±4%	91±3%	86±2%
GradTTS	67±6%	73±5%	79±5%	86±4%	86±4%	78±2%
Comparison to healthy reference						
TransformerTTS	19±6%	21±8%	17±7%	19±8%	18±8%	19±3%
GradTTS	24±8%	25±8%	32±9%	33±9%	30±10%	29±4%
Comparison to pathological reference						
TransformerTTS	13±6%	15±6%	9±5%	11±6%	13±8%	12±3%
GradTTS	17±8%	18±8%	16±8%	15±8%	11±7%	15±4%

outlined in the method, a decline in speaker similarity is observed with increasing speech severity, which aligns with prior findings [10], [11], [35].

When looking at the comparison to the references, the results of the GradTTS agree with our expectation of decreasing severity meaning increased similarity to the healthy reference, and vice versa. We can see a monotonically decreasing trend with the exception of the healthiest (30%, 33%, 32%, 25%, 24%) when comparing to the healthy reference. We can also observe a monotonically increasing trend with the exception of the most severe (11%, 15%, 16%, 18%, 17%) when comparing to the pathological reference.

On the other hand, it is hard to argue that the TransformerTTS has a trend either when comparing to the healthy (18%, 19%, 17%, 21%, 19%) or when comparing to the pathological reference (13%, 11%, 9%, 15%, 13%). We think that this difference can be mostly attributed to the weaker control of severity of the TransformerTTS architecture.

Taking the results together, we conclude both systems are effectively modelling single-speaker, multi-severity speech, where the observed similarity reduction reflects natural variations due to pathology. However, when comparing to the TransformerTTS, we could not show the expected trend of increasing severity, most likely due to the weaker control of the severity.

The effect of severity control is further illustrated by the x-vector cosine similarity distributions in Figure 6. Notably, in the rightmost plot, the TTS Pathological distribution exhibits an almost complete overlap with the Pathological distribution, indicating that the TTS system effectively preserves speaker characteristics in the pathological condition. As severity decreases, the distribution gradually shifts to the right, reflecting a progressive transition toward the control speaker space.

Interestingly, in the TTS Healthy condition, the overlap with the control distribution is incomplete. We hypothesize that this may be due to the imperfect synthesis quality of the healthy samples, which introduces subtle deviations in the x-vector representation. This suggests that while the system successfully modulates severity, some artifacts in the synthesis process may limit its ability to fully reproduce healthy speech characteristics.

E. Limitations of the current work

The model presented in the current work is a single-speaker TTS model, which is a limitation in real-world scenarios, where personalisation would be key. In future work, a cascaded system with text-to-speech synthesis and voice conversion such as [45], or developing a text-to-speech synthesis with separate speaker and severity control could be a way to solve these issues.

While x-vector is one of the best representations that we currently have for pathological speech, we note that it is not perfect, i.e., we qualitatively observe that some severity features are entangled with accented features. In the future, we need to investigate better representations for this task.

Furthermore, another limitation is that the model primarily evaluates and manipulates severity in terms of duration and prosodic variation. This framing, however, omits the critical dimension of articulatory imprecision, which is a primary characteristic of many speech disorders, including those resulting from a glossectomy. For future work, we propose incorporating an explicit articulatory representation into the model. This could potentially be addressed through methods like direct EMA-to-speech synthesis [46] or by using mixed neural-physical models for articulatory synthesis [47]. This would not only strengthen the system's ability to model pathological speech more holistically but also significantly enhance its clinical utility by enabling the simulation of voice outcomes for specific surgical procedures.

VII. CONCLUSION

This paper introduced a new, severity-controllable pathological text-to-speech (TTS) synthesis system. We employed a data augmentation technique for creating a single-speaker dataset with diverse levels of speech severity. We showed that x-vectors contain severity information which can inform the synthesis process. Furthermore, the modified GradTTS architecture had improved duration control, naturalness, stability and intelligibility over the TransformerTTS. We showed that the GradTTS produces more stable, natural and controllable samples than the TransformerTTS. For future work, we would like to investigate how we can disentangle prosody and accent information from the severity information in the x-vector. Furthermore, we would like to develop more stable, and objective measures to evaluate the results of pathological speech synthesis much more reliably, as the connection between naturalness and severity is still a significant challenge.

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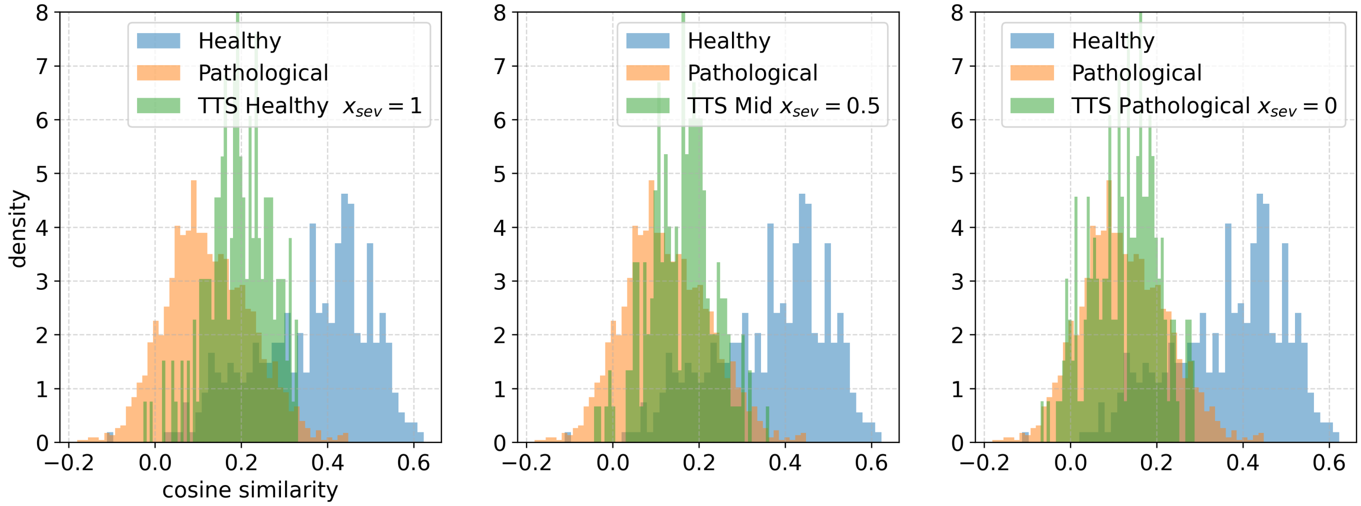

Fig. 6. Comparison of the x-vector distribution of different severity TTS voices to the healthy and pathological references

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